



ISS habitability

2,750Kg of food, fuel and other supplies are on their way to the ISS aboard a Soyuz rocket
<http://dgit.in/foodsyz>



As long as you're inside the protective shield of our magnetosphere, you don't need to worry about the dangerous CMEs. Earth's magnetosphere channels these CMEs around the planet, so that they don't impact us. Without the magnetic field, the solar wind would strip away our atmosphere, like it did to Mars (according to some theories).

The shape of our magnetosphere would be the same as that of a bar magnet's if only the magnetic field of sun was not constantly hitting it. The pressure of solar winds compresses the magnetic fields of the day side of the planet and extends it on the night side giving our magnetosphere the shape of a comet.

Magnetic reconnection

"Magnetic reconnection" is the fundamental process by which magnetic energy is transferred through space. Magnetic reconnection is a process of disconnection and recombination of magnetic field lines. When two loops of magnetic field, with fields of opposite orientation come very close to each other, then it's possible for the lines to break free from their loop and reconnect with the other loop converting the energy stored in the magnetic field to heat and kinetic energy in the form of charged particle acceleration and large-scale flows of matter.

This phenomenon is often seen when the CMEs strike earth's magnetosphere. When the orientation of field lines of plasma is in the opposite direction to that of our magnetosphere, they split and recombine to form complete loops again. This is seen as a great explosion in space, which manifests as spectacular space weather phenomenon such as the Aurora.

NASA's hypothesis

Scientists at NASA believe that the points where the magnetic reconnection occurs

may act as our paths into the next world. But they don't call them portals. They call these points "X-points", "boom tubes" or "electron-diffusion points". NASA believes that the sudden joining of magnetic fields can propel jolts of charged particles from the X points creating an "electron diffusion region".

They're also convinced that these points help in transporting many of the high energy charged particles away from earth.

But, how do they prove that these portals actually exist around earth? The answer is quite simple: by looking at them.

Mission MMS

Launched in March 2015, the Magnetospheric Multiscale (MMS) mission is a Solar Terrestrial Probes Program mission within NASA's Heliophysics Division. MMS investigates how the sun and earth's magnetic fields connect and disconnect, explosively transferring energy from one to the other in magnetic reconnection.

The mission consists of four spacecrafts orbiting the planet in tetrahedron formation. The spacecrafts cover two special places in earth's orbit: one upstream towards the sun and one downstream away from it – both of which places experience magnetic reconnection. The spacecrafts catch the explosions from magnetic reconnection in action by taking high speed snapshots.

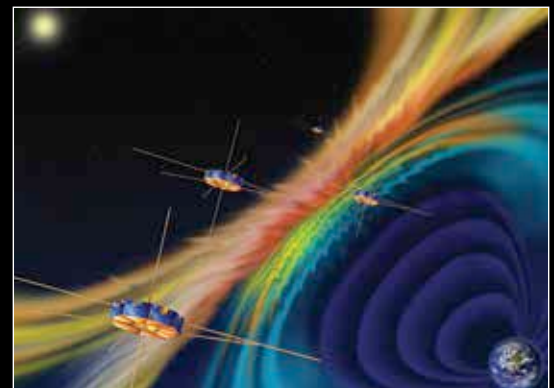
The final step after taking the snapshots is to compare the observations to the simulations of the magnetosphere as it is influenced by the incoming solar wind.

Design and logistics of the mission

The science payload for MMS is all mounted on a single deck plate, but it consists of 25 boxes per spacecraft. These boxes are

categorised into three general groups of instruments: one for measuring fields, one for measuring plasmas and the last for measuring energetic particles. Each spacecraft, along with its detectors, spins around its axis once every 20 seconds. The spacecrafts consist of plasma analysers, energetic particle detectors, magnetometers and electric field instruments as well as a device to prevent spacecraft charging from interfering with the highly sensitive measurements.

The four spacecrafts were launched together on an Atlas V-421 launch vehicle and



The four MMS spacecrafts orbiting Earth in a tetrahedral pattern

were successfully inserted sequentially – at five-minute intervals – into earth's orbit. The spacecrafts are separated by a very small distance of 10 kilometers orbiting earth at a height of 76,000 kilometers.

They are to orbit our planet for two years and six months. Scientists are still waiting for some valuable data to come back from these orbital explorers. The wormholes that they seek, if found, will bring us one step closer to our way into a new unseen universe. Hopefully, we'll find our way around all the paradoxes once we take our first leap of faith through these portals.

To infinity and beyond!

WHITE HOLES?

Certain solutions of the general theory of relativity suggests the presence of a black hole at the mouth of each wormhole. The other end of a wormhole is believed to act in an opposite manner as the black hole (that is, matter and light can escape from it, but can't enter it) and is named the "White Hole".

ARE BLACK HOLES NOT REAL?

After introducing the world to the concept of event horizon and black holes, Stephen Hawking now suggests that the black holes aren't actually real (at least the way we've been thinking of them). He now proposes that there are "apparent horizons" that temporarily suck matter and energy in and eventually releases them out again.

BADLY BEHAVED MAGNETIC FIELD

Neptune, the outermost planet in our solar system, displays some really bizarre behavior which includes hosting a spidery, swirly magnetic field. It's 27 times stronger than the field of earth, sits on an angle with its axis of rotation and is quite asymmetric. After years of study, the nature of this magnetic field is still poorly understood.